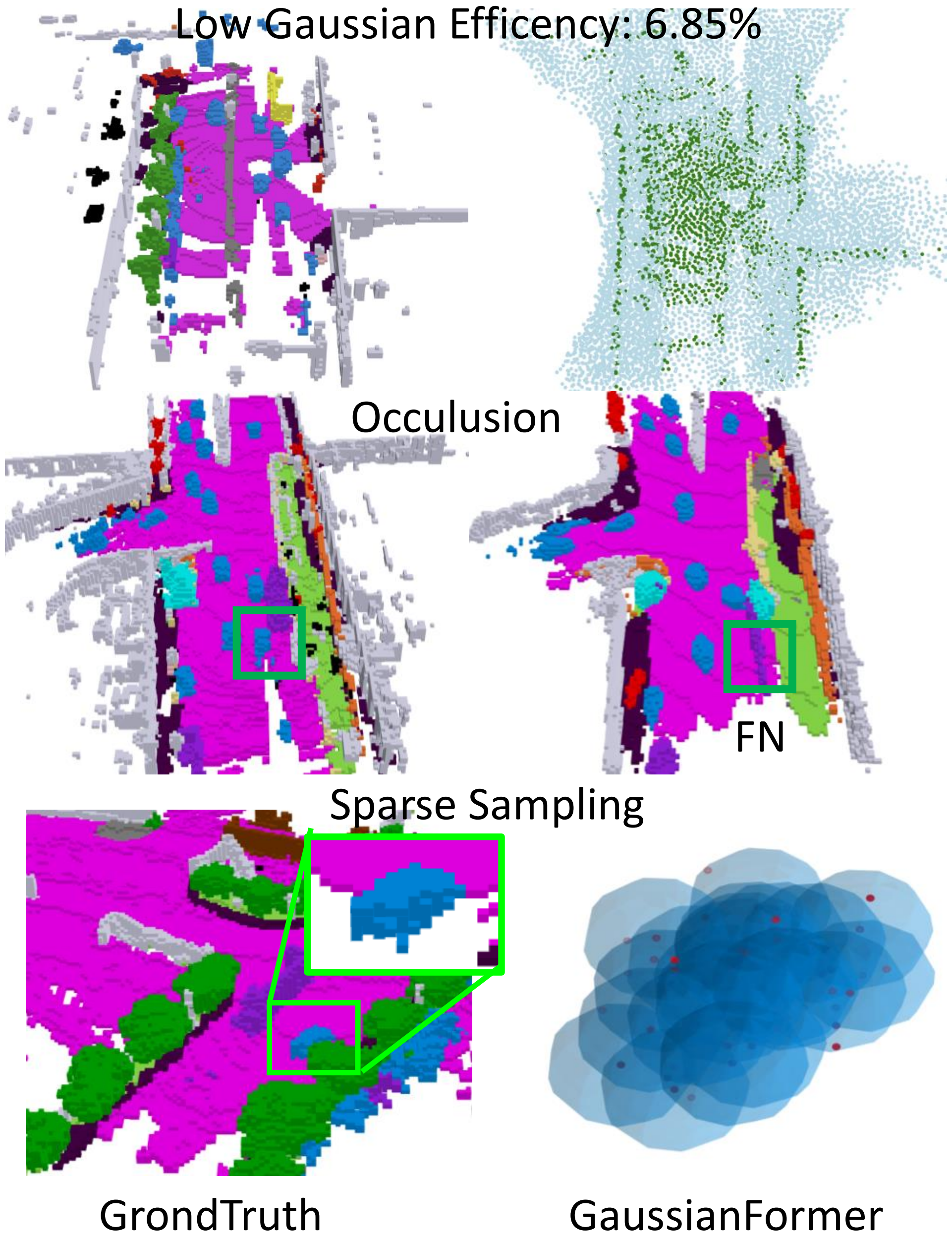
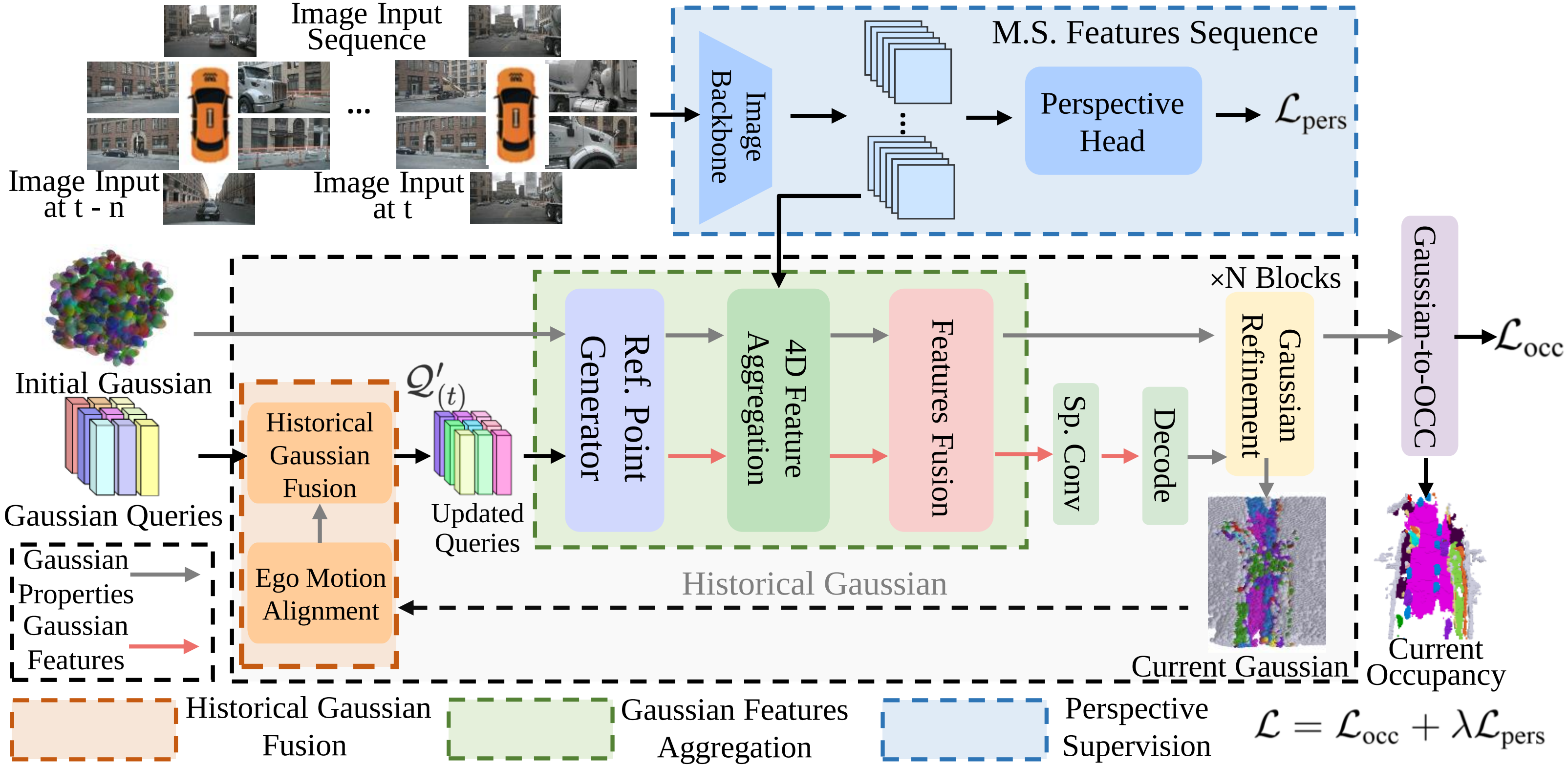
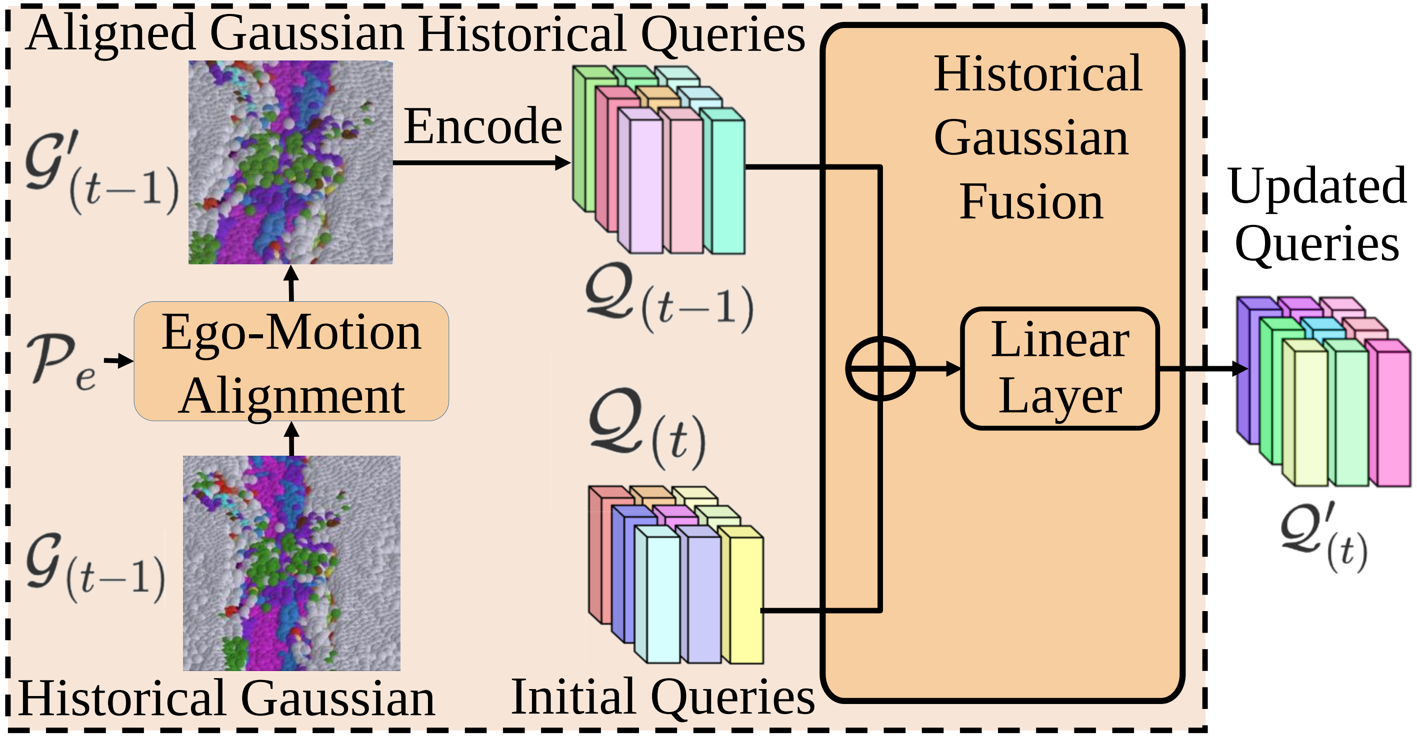
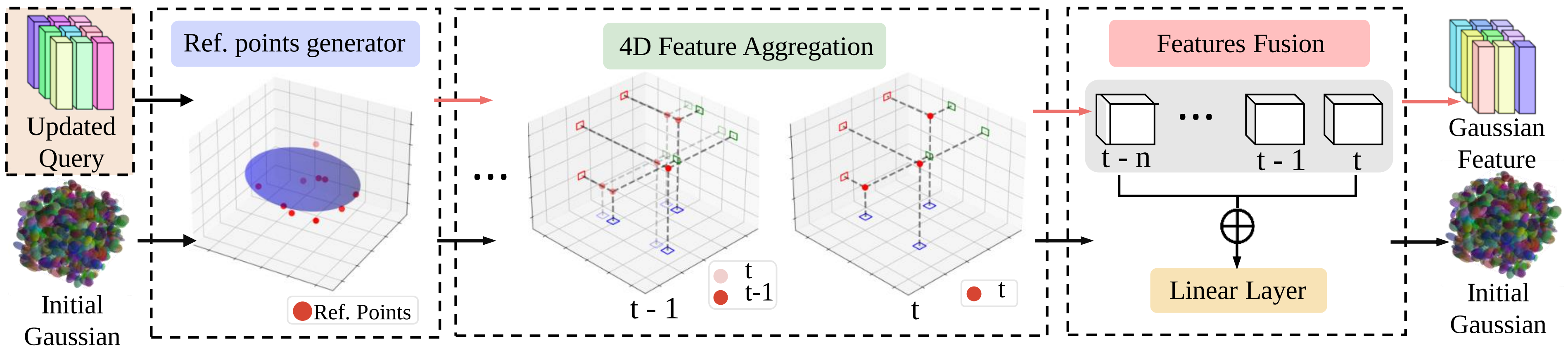

Low Gaussian Efficency: 6.85%
Occulusion
FN
Sparse Sampling
GrondTruth
GaussianFormer



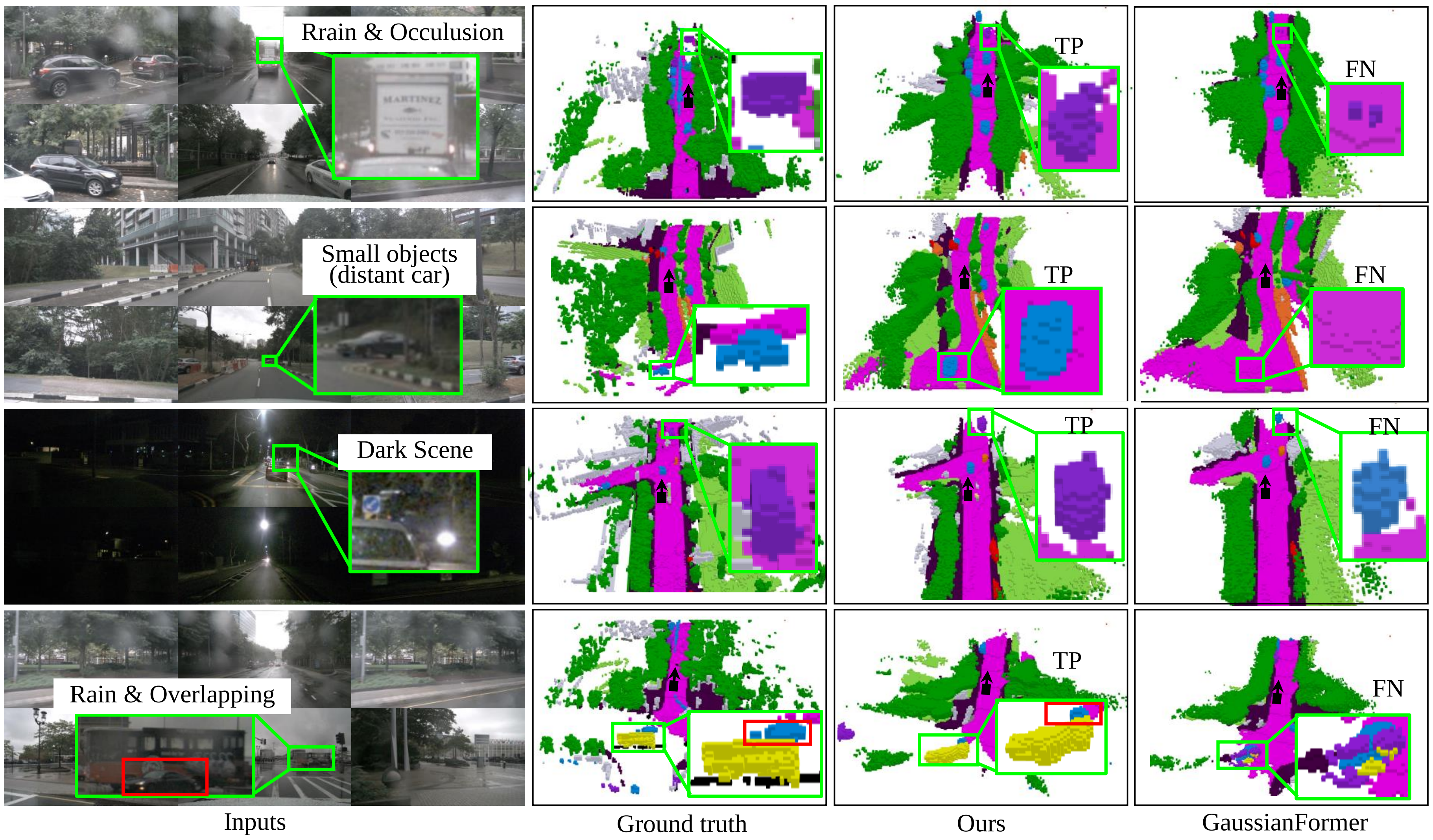
The framework of our Gaussian4DOCC semantic occupancy prediction contains three main parts: 1) historical Gaussian fusion, 2) Gaussian feature aggregation, and 3) perspective supervision.

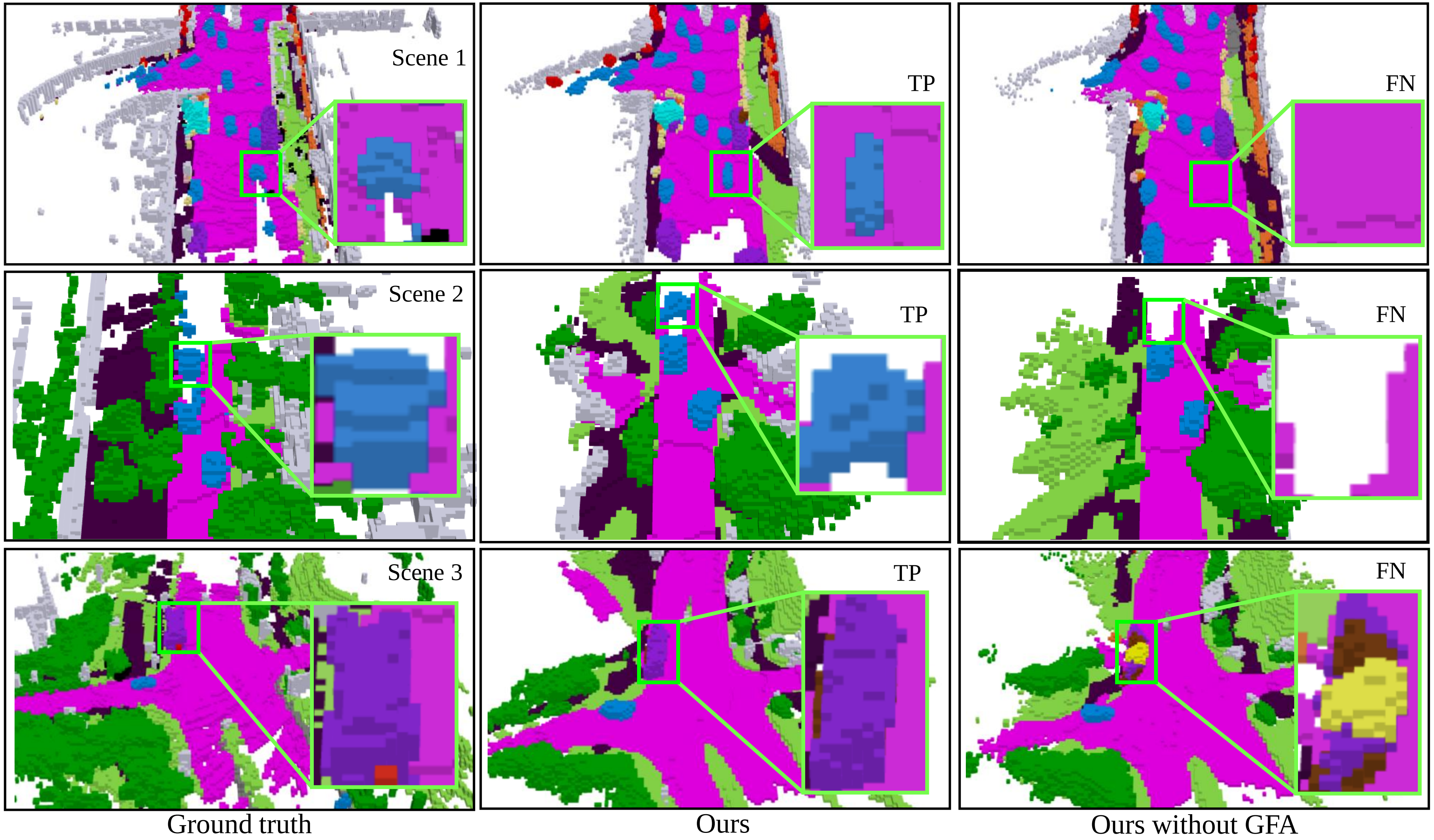


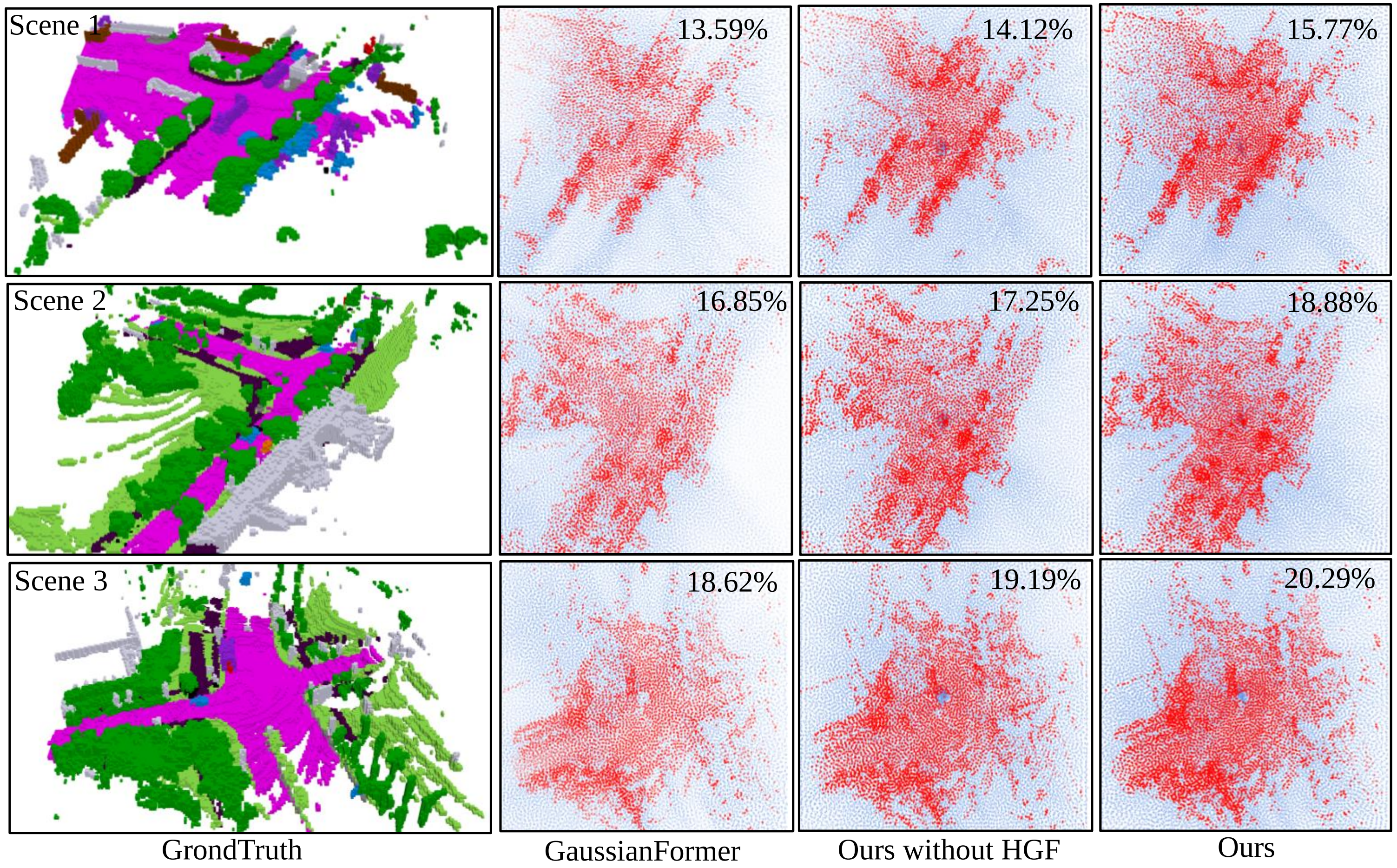
Aligned Gaussian Historical Queries
$\mathcal{G}'_{(t-1)}$
Encode
$\mathcal{Q}_{(t-1)}$
$\mathcal{P}_e$
Ego-Motion Alignment
$\mathcal{Q}_{(t)}$
$\mathcal{G}_{(t-1)}$
Historical Gaussian
Initial Queries
Historical Gaussian Fusion
Linear Layer
Updated Queries
$\mathcal{Q}'_{(t)}$

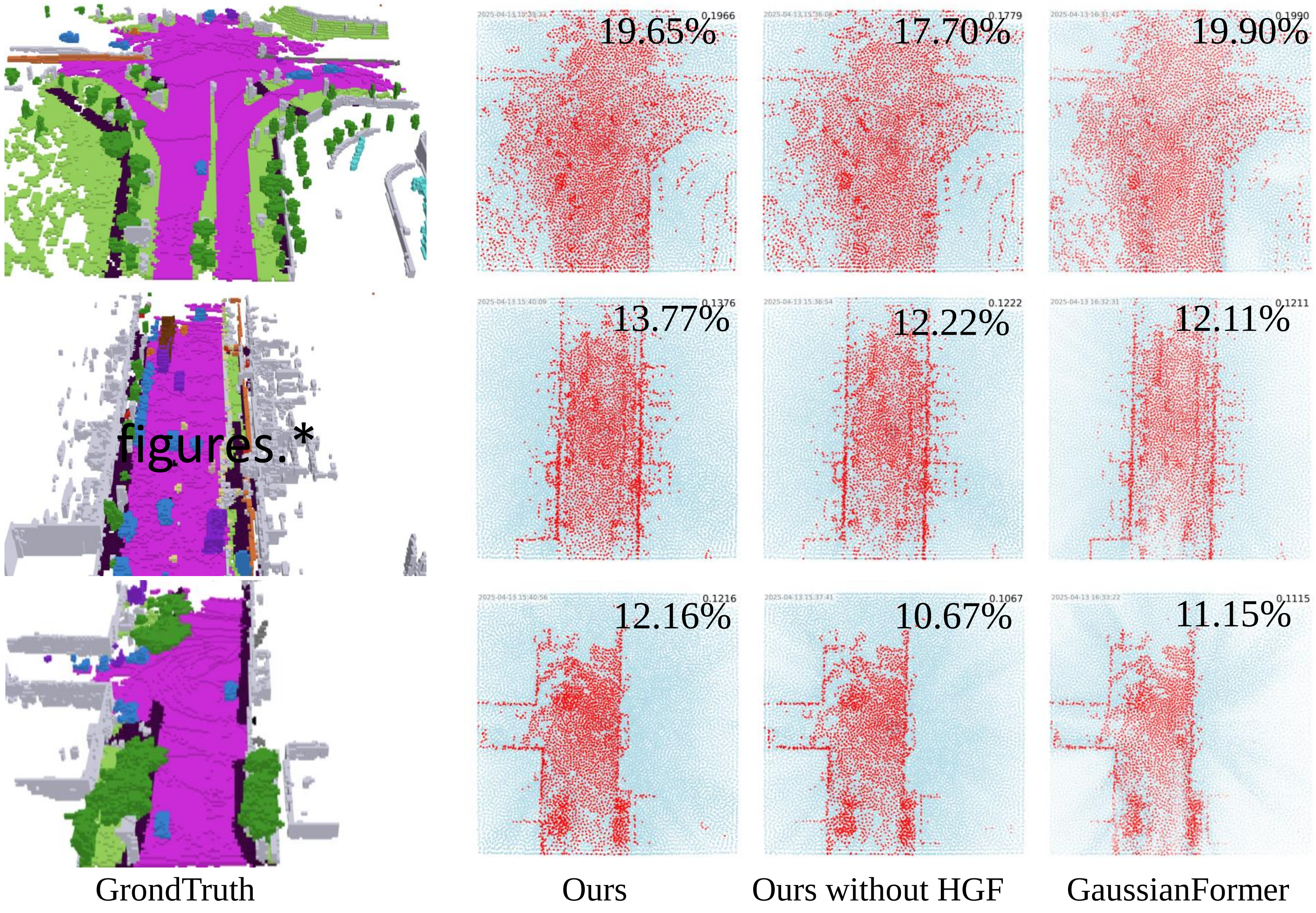


Updated Query
Initial Gaussian
Ref. points generator
Ref. Points
4D Feature Aggregation
t
t-1
t - 1
t
Features Fusion
t - n
t - 1
t
Linear Layer
Gaussian Feature
Initial Gaussian

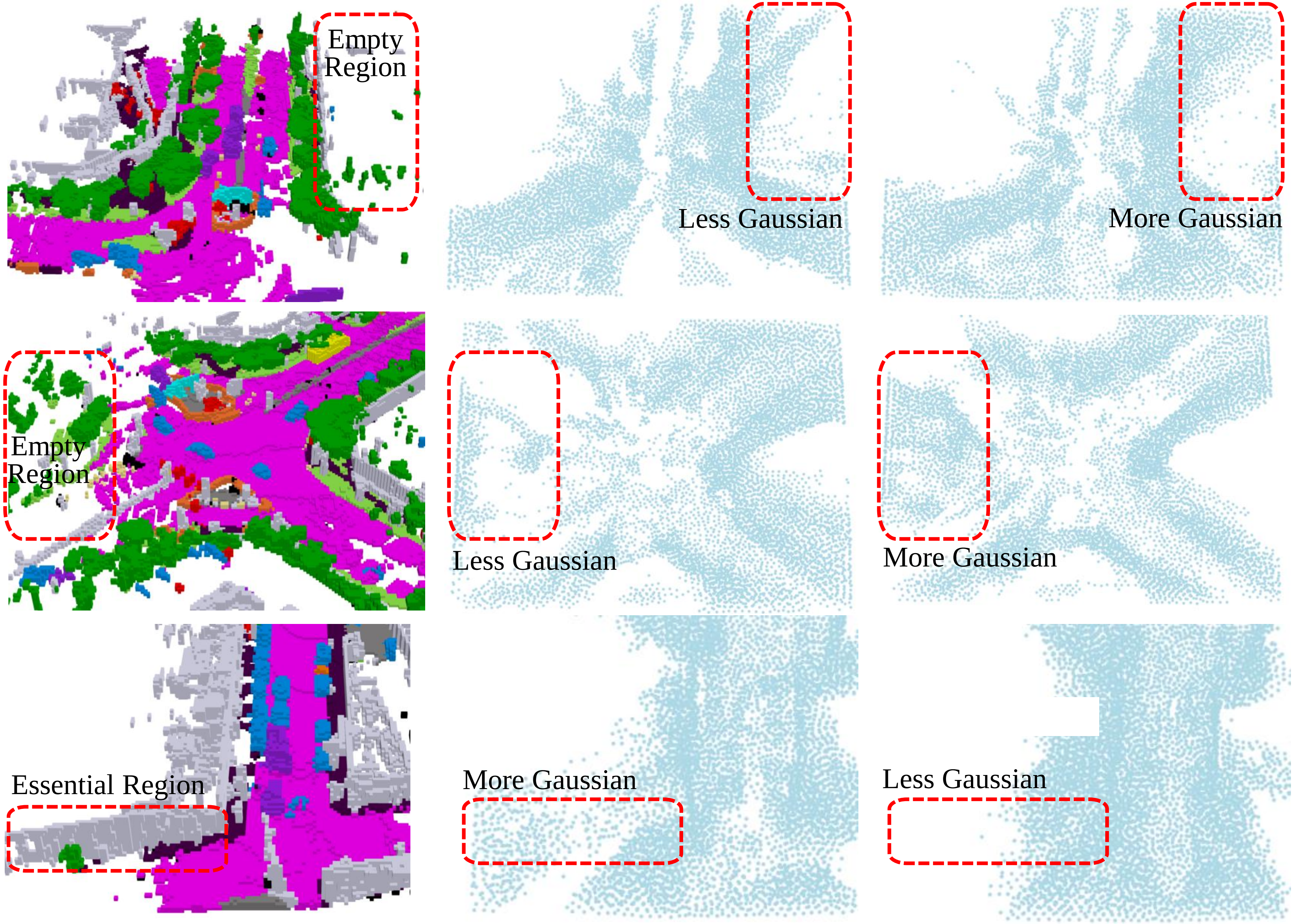

Rrain & Occulusion
Small objects
(distant car)
Dark Scene
Rain & Overlapping
TP
FN
TP
FN
TP
FN
TP
FN
Inputs
Ground truth
Ours
GaussianFormer


Scene 1
TP
FN
Scene 2
TP
FN
Scene 3
TP
FN
Ground truth
Ours
Ours without GFA

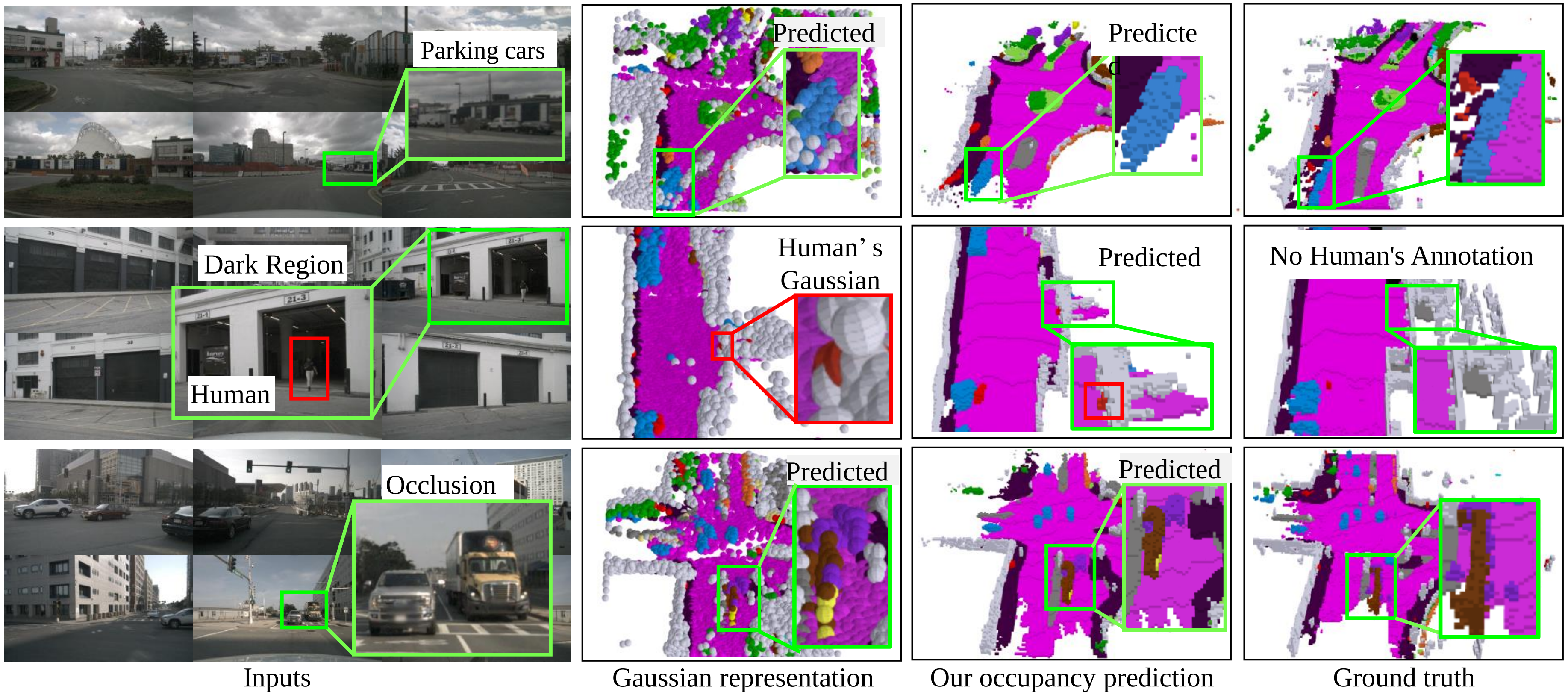

Scene 1
13.59%
14.12%
15.77%
Scene 2
16.85%
17.25%
18.88%
Scene 3
18.62%
19.19%
20.29%
GrondTruth
GaussianFormer
Ours without HGF
Ours



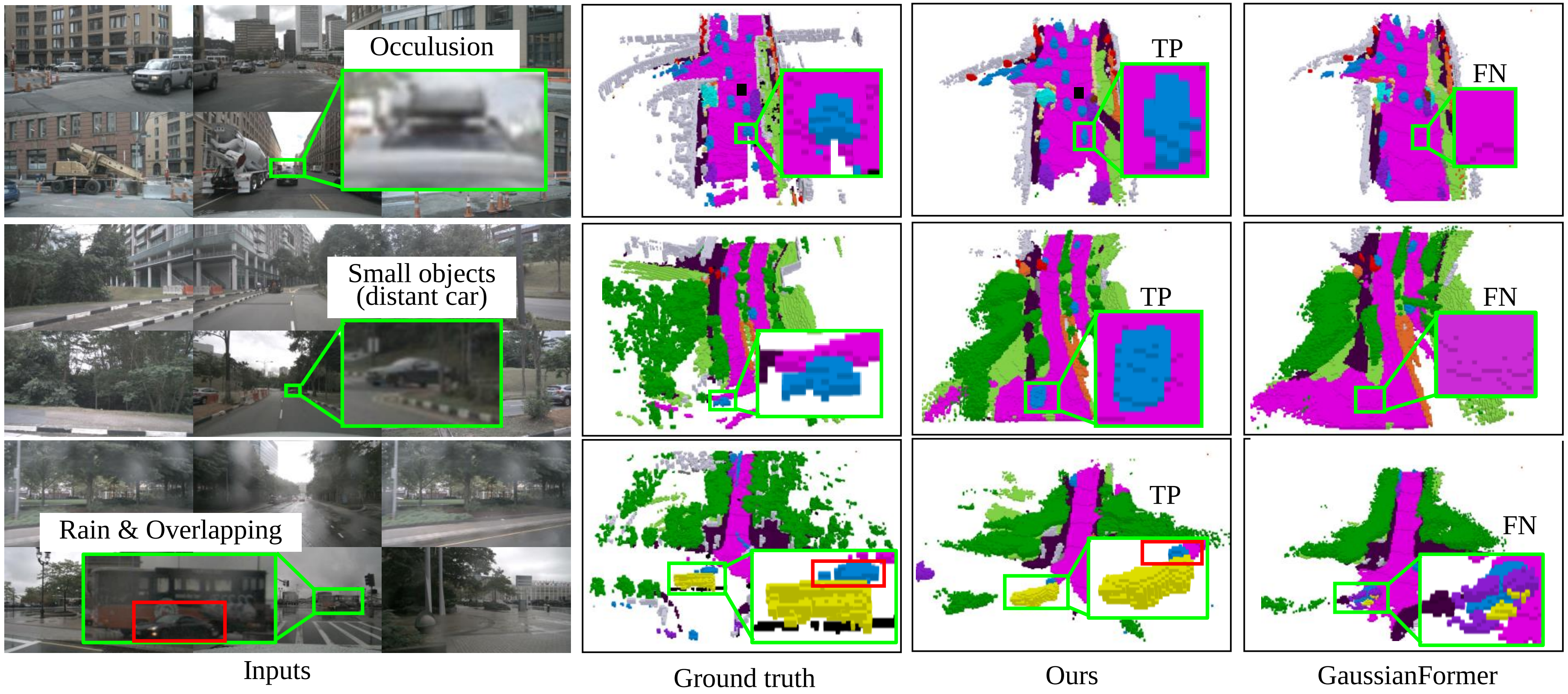
19.65%
17.70%
19.90%
figures.*
13.77%
12.22%
12.11%
12.16%
10.67%
11.15%
GrondTruth
Ours
Ours without HGF
GaussianFormer

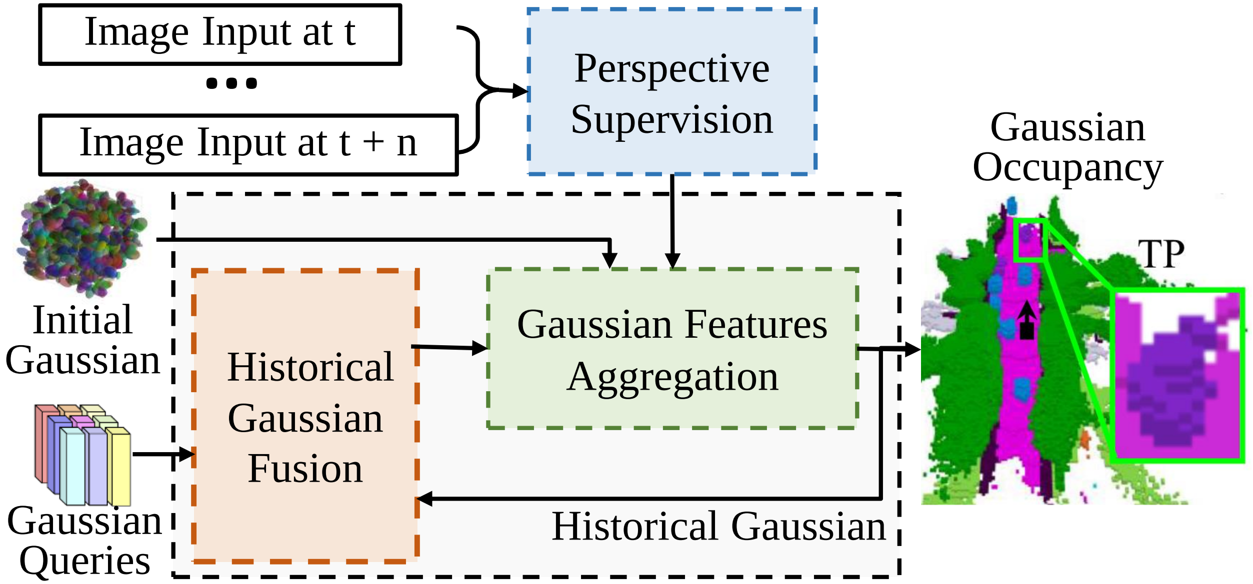


Empty
Region
Less Gaussian
More Gaussian
Empty
Region
Less Gaussian
More Gaussian
Essential Region
More Gaussian
Less Gaussian

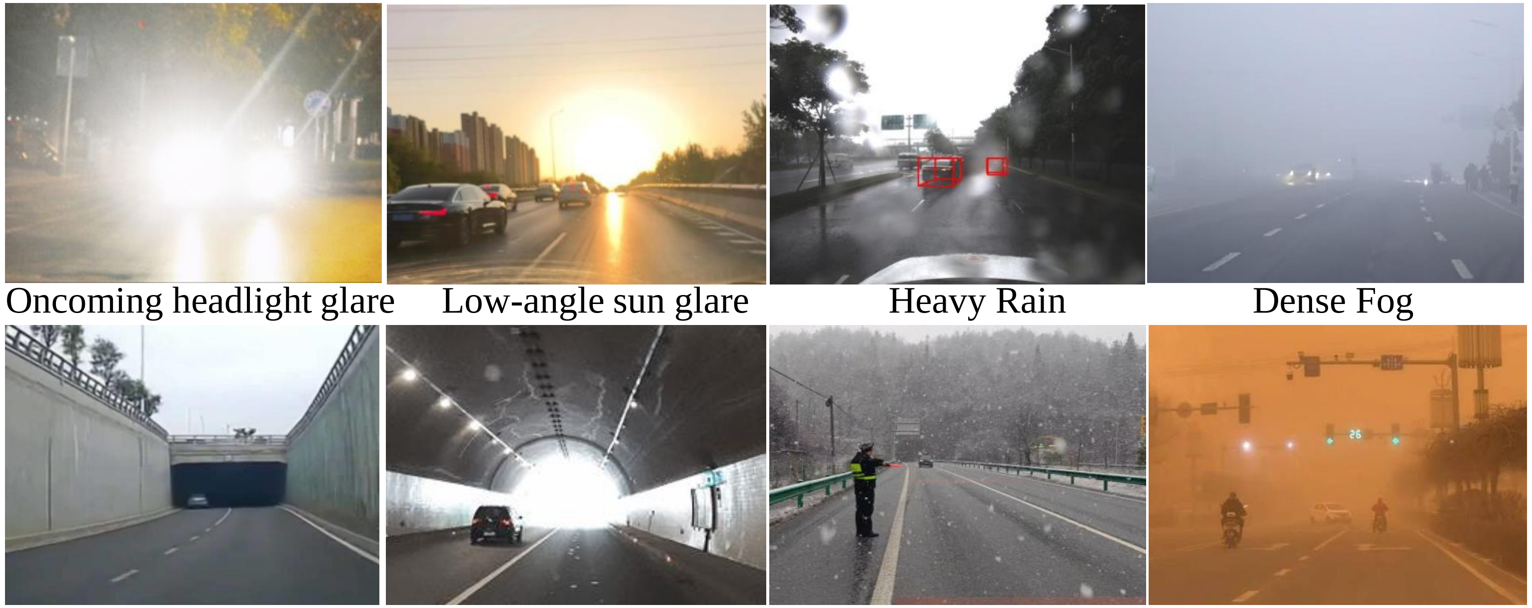


Parking cars
Dark Region
Human
Occlusion
Predicted
Predicte
d
Human' s
Gaussian
Predicted
No Human's Annotation
Predicted
Predicted
Inputs
Gaussian representation
Our occupancy prediction
Ground truth


Occulusion
TP
FN
Small objects
(distant car)
TP
FN
Rain & Overlapping
TP
FN
Inputs
Ground truth
Ours
GaussianFormer



Our Gaussian4DOCC contains three contributions:
1) historical Gaussian fusion, 2) Gaussian feature aggregation,
3) perspective supervision



Oncoming headlight glare | Low-angle sun glare | Heavy Rain | Dense Fog

Black hole effect | White hole effect | Heavy Snow | Dust Storm